Longest Arithmetic Subsequence (medium)

Problem Statement

Given an array `nums` containing positive integers, return the maximum length of a subsequence that forms an `arithmetic progression`.

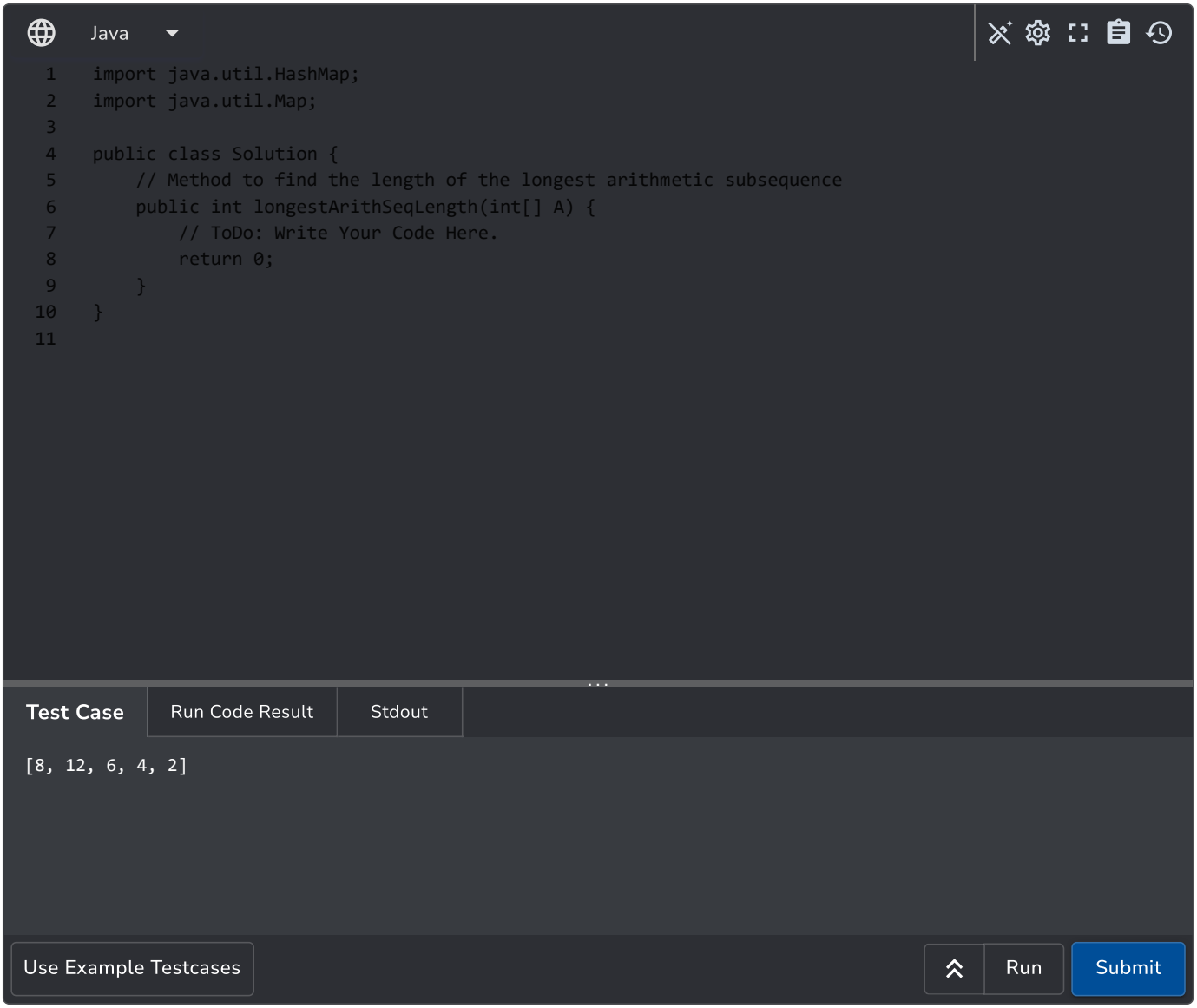
- A `subsequence` is an array that can be formed from `nums` by deleting 0 or more elements without changing the order of the remaining elements.
- An arithmetic progression (AP) is a sequence of numbers in which the difference between any two consecutive elements is constant. In short, `seq[i + 1] - seq[i]` should be same for all `0 <= i < seq.length - 2`.

Examples

- **Example 1**:
  - **Input**: `[8, 12, 6, 4, 2]`
  - **Expected Output**: `4`
  - **Justification**: The subsequence `[8, 6, 4, 2]` forms the longest arithmetic subsequence with a common difference of **-2**, thus the longest arithmetic subsequence has a length **4.**
- **Example 2**:
  - **Input**: `[5, 10, 15, 7, 11, 5]`
  - ****Expected Output: `3`
  - Justification:** The subsequence [**5, 10, 15**] forms the longest arithmetic subsequence with a common difference of **5.**
- **Example 3**:
  - **Input**: `[1, 7, 10, 15, 27, 29]`
  - **Expected Output**: `3`
  - **Justification**: The subsequence [**1, 15, 29**] forms the longest arithmetic subsequence with a common difference of **14.**

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Solution: Longest Arithmetic Subsequence

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